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BHADARWAH CAMPUS

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(PRACTICAL FILE)

PYTHON PROGRAMMING

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Thank you.

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CERTIFICATE

This is to certify that **VIKAS KUMAR** student of **BCA-4th semester** bearing university **Roll no. 2** has completed the required number of practical of under the guidance of class teacher **Mr. ABHAY KHAJURIA SIR.** from the Department of Computer science and IT, Bhaderwah campus, University of Jammu.

Signature

(I/C COURSE COORDINATOR)

Signature

(PRACTICAL I/C)

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PRACTICAL 1.

```
print("Hello world ! ")
```

OUTPUT :

```
PS C:\Users\mca\Desktop\python> python -u "c:\Users\mca\Desktop\python\.vscode\HELLO WORLD .PY"
PS C:\Users\mca\Desktop\python> python -u "c:\Users\mca\Desktop\python\.vscode\HELLO WORLD .PY"
PS C:\Users\mca\Desktop\python> python -u "c:\Users\mca\Desktop\python\.vscode\HELLO WORLD .PY"
Hello world !
PS C:\Users\mca\Desktop\python> python -u "c:\Users\mca\Desktop\python\.vscode\HELLO WORLD .PY"
Hello world !
PS C:\Users\mca\Desktop\python> python -u "c:\Users\mca\Desktop\python\.vscode\HELLO WORLD .PY"
Hello world !
PS C:\Users\mca\Desktop\python> python -u "c:\Users\mca\Desktop\python\.vscode\HELLO WORLD .PY"
Hello world !
PS C:\Users\mca\Desktop\python> python -u "c:\Users\mca\Desktop\python\.vscode\HELLO WORLD .PY"
Hello world !
PS C:\Users\mca\Desktop\python> python -u "c:\Users\mca\Desktop\python\.vscode\HELLO WORLD .PY"
Hello world !
PS C:\Users\mca\Desktop\python> python -u "c:\Users\mca\Desktop\python\.vscode\HELLO WORLD .PY"
Hello world !
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 2 :

```
num=int(input("Enter a Number:"))

if num >0:

    print("num is positive")

else:

    print("num is negative")
```

OUTPUT :

```
Enter a Number:77
num is positive
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 3 :

```
num=int(input("Enter a number:"))
if (num % 2)==0:
 print("GIVEN NUMBER IS EVEN")
else:
   print("GIVEN NUMBER IS ODD")
```

OUTPUT :

1).

```
Enter a number:6
GIVEN NUMBER IS EVEN
```

2).

```
Enter a number:7
GIVEN NUMBER IS ODD
```

PRACTICAL 4 :

```
num=int(input("Enter a Number:"))
if num>0 and num%2==0:
   print("Number Is Positive And Even")
else:
   print("Number Is Not Positive And Even")
```

OUTPUT :

1).

```
Enter a Number:6
Number Is Positive And Even
```

2).

```
Enter a Number:-6
Number Is Not Positive And Even
```

```
Enter a Number:7
Number Is Not Positive And Even
```

PRACTICAL 5 :

```
for i in range(1,8):
    print(i)
```

OUTPUT :

```
1
2
3
4
5
6
7
```

PRACTICAL 6 :

```
for x in range(10,0,-1):
    print(x)
```

OUTPUT :

```
10
9
8
7
6
5
4
3
2
1
```

PRACTICAL 7 :

```
i=0
while i<7:
    i+=1
    print(i)
```

OUTPUT:

```
1
2
3
4
5
6
7
```

PRACTICAL 8:

CONTINUE:

```
i=0
while i<9:
    i+=1
    if i==7:
        continue
    print(i)
```

OUTPUT:

```
1
2
3
4
5
6
8
9
```

BREAK :

```
i=0
while i <=17:
    print(i)
    if i==6:
        break
    i+=1
```

OUTPUT:

```
0
1
2
3
4
5
6
```

PRACTICAL 9:

```
num= int(input("Enter  marks:"))

if num >=90:

   print("Grade = A+")

elif num >=80:

   print("Grade  = A ")

elif num >=70:

   print("Grade = B+ ")

elif num >= 60:

   print("Grade = B ")

elif num >= 47:

   print("Grade = Good")

else:

   print("fail")
```

OUTPUT:

```
Enter marks:77
Grade = B+
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 10:

```
print("simple calculator")
while True:
    print("\nChoose operation:")
    print("1.Add")
    print("2.Subtract")
    print("3.multiply")
    print("4.Divide")
    print("5.Exit")


    choice = input("Enter choice (1-5):")
    if choice =="5":
        print("calculator closed")
        break

    num1=float(input("Enter first number:"))
```

```
num2=float(input("Enter second number:"))


if choice == "1":
    print("Result:",num1+num2)


elif choice == "2":
        print("Result:",num1 - num2)
elif choice == "3":
            print("Result:",num1*num2)
elif choice == "4":
    if num2!=0:
        print("Result:",num1/num2)
    else:
        print("cannot divide by zero!")
else:
        print("invalid choice!")
```

OUTPUT:

```
Choose operation:
1.Add
2.Subtract
3.multiply
4.Divide
5.Exit
Enter choice (1-5):3
Enter first number :11
Enter second number :7
Result: 77.0

Choose operation:
1.Add
2.Subtract
3.multiply
4.Divide
5.Exit
Enter choice (1-5):
```

CLOSED CALCULATOR:

```
Choose operation:
1.Add
2.Subtract
3.multiply
4.Divide
5.Exit
Enter choice (1-5):5
calculator closed
PS C:\Users\mca\Desktop\python> 5
5
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 11:

```
num1 = 10          #integer
num2  = 3.27       #float
text = "Python"    # string
print("integer :", num1)
print("Float value :", num2)
print("String :", text)
```

OUTPUT:

```
integer : 10
Float value : 3.27
String : Python
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 12:

```
str1 = "Hello"

str2 = "world"


#concatenation

result = str1 + " " + str2

print("string concatenation and replication")

print("concatenation :", result)

#replication

print("replication:",str1 * 7)
```

OUTPUT:

```
string concatenation and replication
concatenation : Hello world
replication: HelloHelloHelloHelloHelloHelloHello
```

PRACTICAL 13:

```
Name = "VIKAS KUMAR"

Age = 20

print("Name :", Name)

print("Age :", Age)
```

OUTPUT:

```
Name : VIKAS KUMAR
Age : 20
```

PRACTICAL 14:

```
Name = input("Enter your name :")

Age = int(input("Enter your age : "))

print("Name : ", Name)

print("Age : ", Age)
```

OUTPUT:

```
Enter your name :VIKAS KUMAR
Enter your age : 20
Name :  VIKAS KUMAR
Age :  20
```

PRACTICAL 15:

```
A = int(input("Enter first number : "))

B = int(input("Enter second number : "))

print("Before swapping : ", A,B)

C = A

A = B

B = C

print("After swapping :", A,B)
```

OUTPUT:

```
Enter first number : 7
Enter second number : 8
Before swapping :  7 8
After swapping : 8 7
```

PRACTICAL 16:

```
A = 25

B = 27

print("A!= : ", A!=B)

print("A==B : ", A==B)

print("A>B :", A>B)

print("A<B :", A<B)
```

OUTPUT:

```
PS C:\Users\mca\Desktop\python> python -u "c:\Users\mca\Desktop\python\.vscode\BOOLEAN VALUES AND COMPARSION.PY"
A!= :  True
A==B :  False
A>B : False
A<B : True
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 17:

```
x = True

y = False


print("Boolean operators : ")

print("AND :", x and y)

print("OR : ", x or y)

print("NOT : ", not x)
```

OUTPUT:

```
Boolean operators :
AND : False
OR :  True
NOT :  False
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 18:

```
age = int(input("Enter your AGE : "))

has_id = True

if age >=18 and has_id:

    print("You can vote ")

else:

    print("You are not eligible")
```

OUTPUT:

```
Enter your AGE : 19
You can vote
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 19:

```
a = int(input("Enter first number : "))

b = int(input("Enter second number : "))

c = int(input("Enter Third number :  "))


if a > b and a > c:

    print("Largest number is : ", a)
```

```
elif b > c:
    print("Largest number is : ", b)
else:
    print("Largest number is : ", c)
```

OUTPUT:

```
Enter first number : 27
Enter second number : 6
Enter Third number :  7
Largest number is :  27
```

PRACTICAL 20:

```
import math
print("square root of 49 : ", math.sqrt(49))
```

OUTPUT:

```
square root of 49 :  7.0
```

PRACTICAL 21:

```
import sys

print("start")

sys.exit()

print("This will not print")
```

OUTPUT:

```
PS C:\Users\mca\Desktop\python> p
start
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 22:

```
def add(a,b):
 return a+b
result=add(7,70)
print("RESULT OF FUNCTION")
print(result)
```

OUTPUT:

```
RESULT OF FUNCTION
77
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 23:

```
def info(name,age):
   print(name,age)
info(age=20, name="VIKAS KUMAR")
```

OUTPUT:

```
VIKAS KUMAR 20
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 24:

```
x =27   #Global

def test():

    x =5   #Local

    print("local x :", x)


test()

print("Global x : ", x)
```

OUTPUT:

```
local x : 5
Global x :  27
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 25:

```
try:

   num = int(input("Enter number: "))

   result = 10 / num

   print("Result:", result)

except ValueError:

   print("Invalid input")

except ZeroDivisionError:
```

```
    print("Cannot divide by zero")
else:
    print("No error")
finally:
    print("Program ended")
```

OUTPUT:

```
Enter number: 7
Result: 1.4285714285714286
No error
Program ended
```

```
Enter number: 0
Cannot divide by zero
Program ended
```

```
Enter number: 3.14
Invalid input
Program ended
```

PRACTICAL 26:

```
fruits = ["apple", "banana", "mango"]
print(fruits)
```

OUTPUT:

```
['apple', 'banana', 'mango']
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 27:

```
numbers = [1, 2, 3, 4]

numbers.append(5)

numbers.remove(2)

print(numbers)
```

OUTPUT:

```
[1, 3, 4, 5]
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 28:

```
nums = [5, 2, 9, 1]

nums.sort()

print(nums)

nums.reverse()

print(nums)
```

OUTPUT:

```
[1, 2, 5, 9]
[9, 5, 2, 1]
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 29:

```
student = {
    "name": "Vikas",
    "age": 20,
    "course": "Python"
}
print("Name:", student["name"])
print("Age:", student["age"])
```

OUTPUT:

```
Name: Vikas
Age: 20
PS C:\Users\mca\Desktop\python>
```

PRACTICAL 30:

```
# Dictionary
... student = {
...     "name": "Vikas",
...     "age": 20,
...     "marks": [90, 85, 88]
... }
```

```
...
... # Pretty print
... pprint.pprint(student)
...
... # Access data
... print("Name:", student["name"])
...
... # Loop
... for key, value in student.items():
...     print(key, ":", value)
...
... # String operations
... msg = "hello python"
...
... print(msg.upper())
... print(msg.replace("python", "world"))
```

OUTPUT:

```
{'age': 20, 'marks': [90, 85, 88], 'name': 'Vikas'}
Name: Vikas
name : Vikas
age : 20
marks : [90, 85, 88]
HELLO PYTHON
hello world
>>>
```

PRACTICAL 31:

```
>>> msg = "hello python"
...
... print(msg.upper())
... print(msg.lower())
... print(msg.replace("python", "world"))
... print(msg.split())
```

OUTPUT:

```
HELLO PYTHON
hello python
hello world
['hello', 'python']
>>>
```

PRACTICAL 32:

```
>>> username = input("Enter username: ")
... password = input("Enter password: ")
...
... if username == "VIKAS KUMAR":
```

```
...     if password == "1234567":
...         print("Login successful")
...     else:
...         print("Wrong password")
... else:
...     print("Invalid username")
```

OUTPUT:

```
Enter username: VIKAS KUMAR
Enter password: 1234567
Login successful
>>> |
```

PRACTICAL 33:

```
import sys


# Account Data (Dictionary)
account = {
    "name": "Vikas",
    "pin": "1234567",
    "balance":70000
}


transactions = []  # List to store history



# Login Function
def login():
    entered_pin = input("Enter your PIN: ").strip()
    if entered_pin == account["pin"]:
        print("Login Successful!\n")
        return True
    else:
        print("Wrong PIN!\n")
        return False
```

```
# Check Balance
def check_balance():
    print("Current Balance:", account["balance"], "\n")


# Deposit Money
def deposit():
    try:
        amount = float(input("Enter amount to deposit: "))
        account["balance"] += amount
        transactions.append(f"Deposited {amount}")
        print("Deposit successful!\n")
    except ValueError:
        print("Invalid amount!\n")


# Withdraw Money
def withdraw():
    try:
        amount = float(input("Enter amount to withdraw: "))
        if amount <= account["balance"]:
            account["balance"] -= amount
            transactions.append(f"Withdrew {amount}")
```

```
            print("Withdrawal successful!\n")
        else:
            print("Insufficient balance!\n")
    except ValueError:
        print("Invalid amount!\n")



# View Transactions
def view_transactions():
    if len(transactions) == 0:
        print("No transactions yet.\n")
    else:
        print("Transaction History:")
        for t in transactions:
            print("-", t)
        print()



# Main Menu
def main():
    if not login():
        sys.exit()

    while True:
```

```
print("===== ATM MENU =====")
print("1. Check Balance")
print("2. Deposit")
print("3. Withdraw")
print("4. Transaction History")
print("5. Exit")

choice = input("Enter choice: ")

if choice == "1":
    check_balance()
elif choice == "2":
    deposit()
elif choice == "3":
    withdraw()
elif choice == "4":
    view_transactions()
elif choice == "5":
    print("Thank you for using ATM!")
    sys.exit()
else:
    print("Invalid choice!\n")
```

```
# Run Program

main()
```

OUTPUT:

1. **Login using Pin:**

```
Enter your PIN: 1234567
Login Successful!

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice:
```

2. **Check balance:**

```
Enter your PIN: 1234567
Login Successful!

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 1
Current Balance: 70000
```

3. **Deposit money:**

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 2
Enter amount to deposit: 700
Deposit successful!
```

4. **Check Balance after deposit money:**

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 1
Current Balance: 70700.0
```

5. **Withdraw money:**

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 3
Enter amount to withdraw: 7077
Withdrawal successful!
```

6. **Check balance after withdrawal money:**

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 1
Current Balance: 63623.0
```

7. **Exit program:**

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 5
Thank you for using ATM!
```

8. **ALL OUTPUT:**

```
Enter your PIN: 1234567
Login Successful!

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 1
Current Balance: 70000

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 2
Enter amount to deposit: 700
Deposit successful!

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 1
Current Balance: 70700.0

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 3
Enter amount to withdraw: 7077
Withdrawal successful!
```

```
===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 1
Current Balance: 63623.0

===== ATM MENU =====
1. Check Balance
2. Deposit
3. Withdraw
4. Transaction History
5. Exit
Enter choice: 5
Thank you for using ATM!
```

THANK YOU SIR